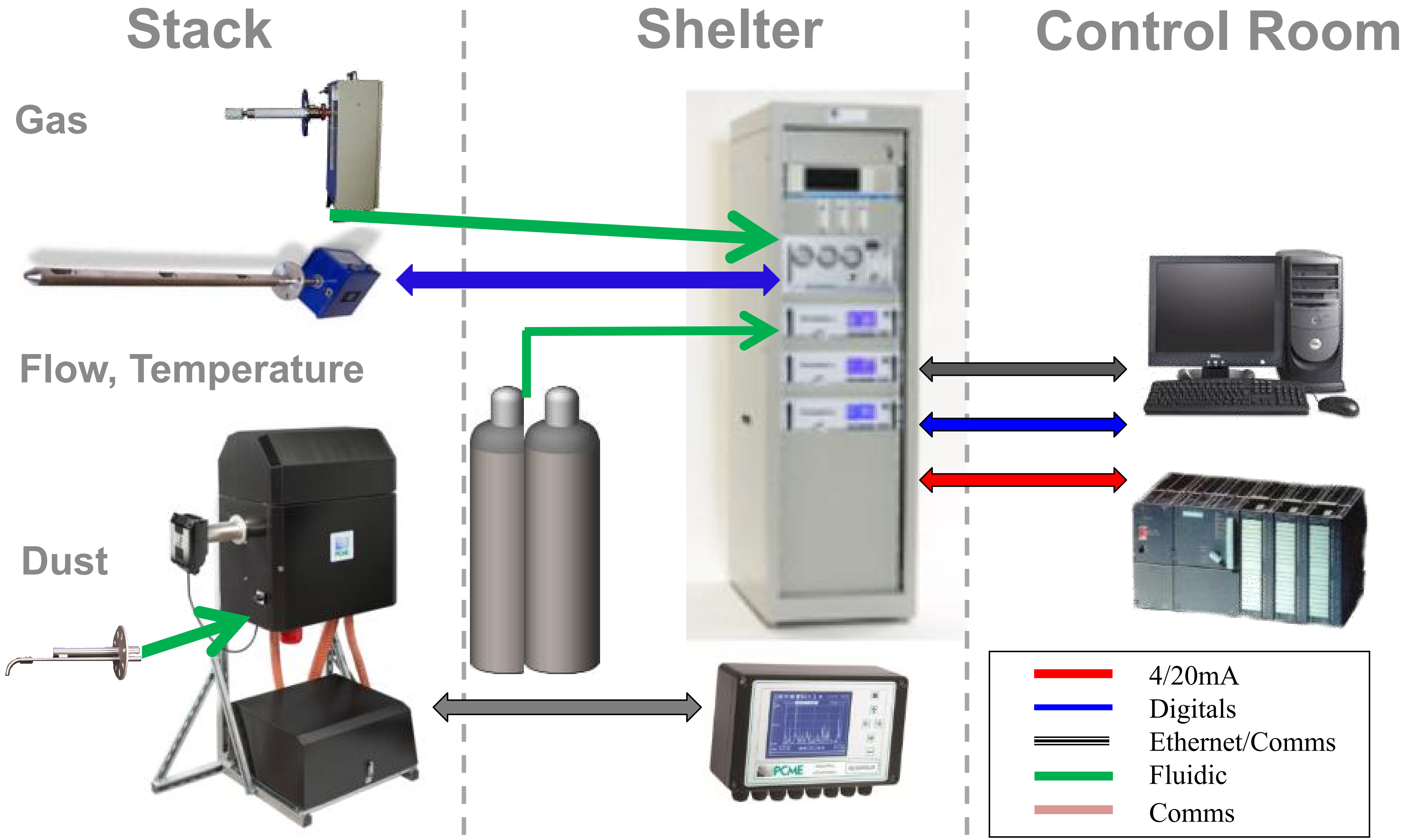
Typical CEMS Overview



Why do our valued customers install particulate systems?

- To comply with Legislation.
- To comply with their Authorisation or Permit.
- To optimise their process
  - Increase production by reducing loss of product through the filters/dryers.
  - Gain better knowledge of process.
  - Improve and reduce maintenance.
  - Save on cost of labour and filter media.
- To comply with corporate requirements:

Or for improved protection of employees.

Introduction to Particulate Monitoring

With the implementation:

- In US of MACT/MAT (Europe the Industrial Emission Directive (IED)), and performance standards such as PS-11 (EN14181).

There is an increasing requirement of more stringent and lower Emission Limit Values and as such new abatement techniques are being implemented and control of emissions need to be demonstrated by continuous measurement and reporting.

One of the common parameters that needs to be measured and reported on a continuous basis is Particulate (mg/m3).

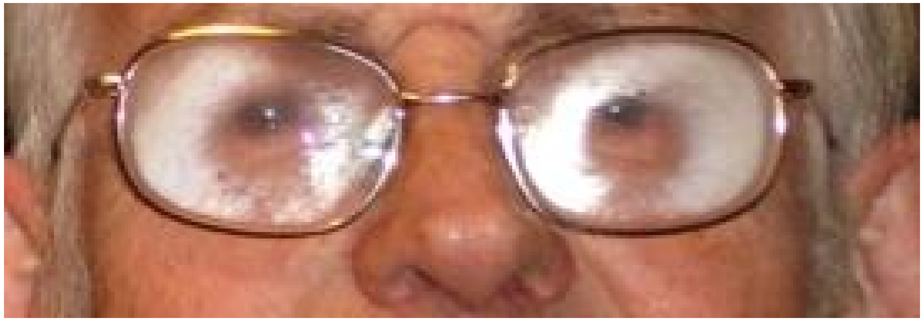
Typically in dry stacks applications the most efficient and cost effective method is to measure Particulate directly in the process itself (in situ).

And for Wet stacks an extractive system is required that removes the water droplets before measurement.

Is your process dry, humid or wet?

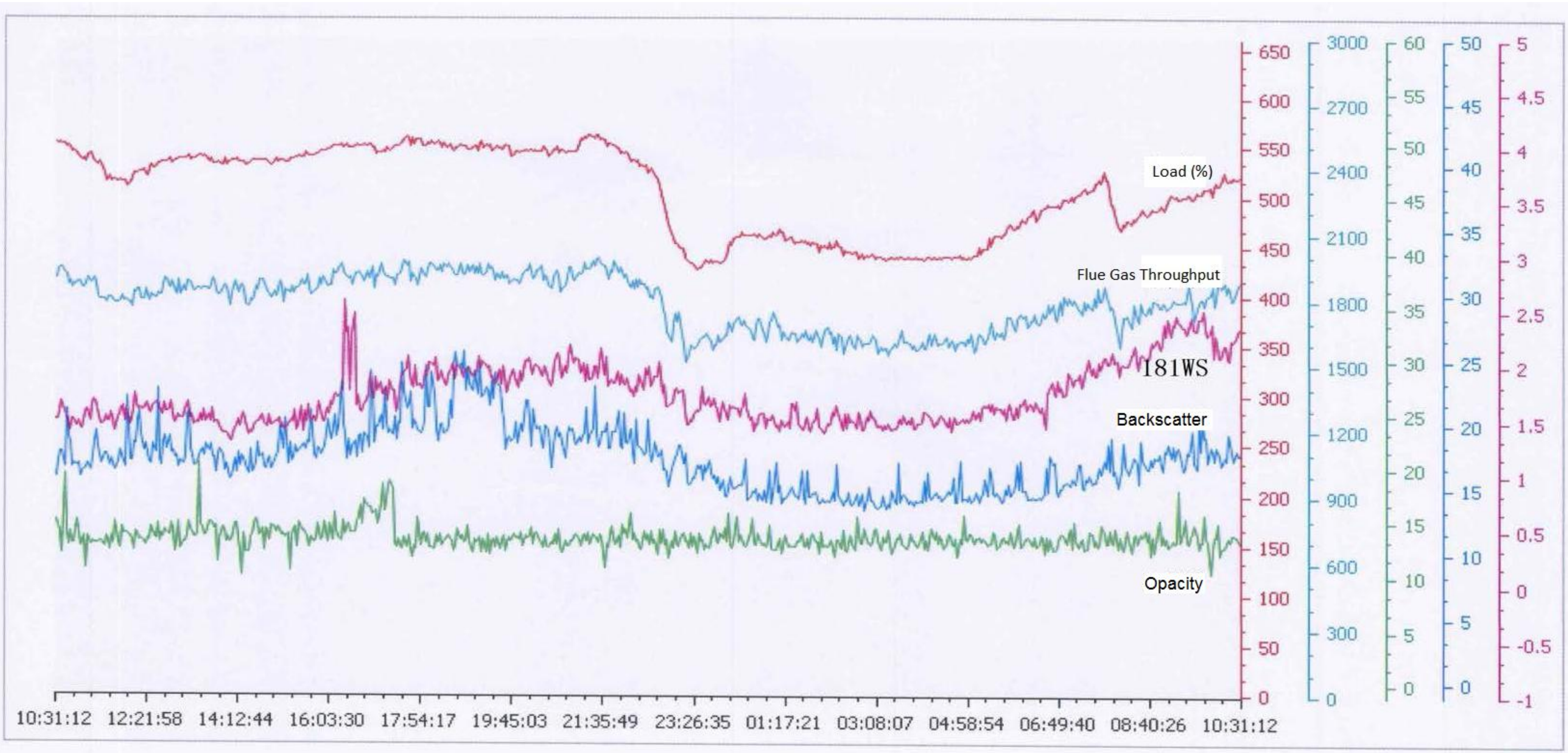
- ✓ Dry-The dust tends not stick easily to probe based systems so air purge is not required except on optical based systems.

- ✓ Humid-The dust will stick to probe and optical based systems so air purge is required or insulated probes can be considered.



- ✓ Wet-The water droplets will be interfering on all in-situ dust measurement systems so extractive dust monitoring system will be required.

Wet System vs Dry Systems



Points to consider when choosing what Particulate Monitoring System is suitable for you?

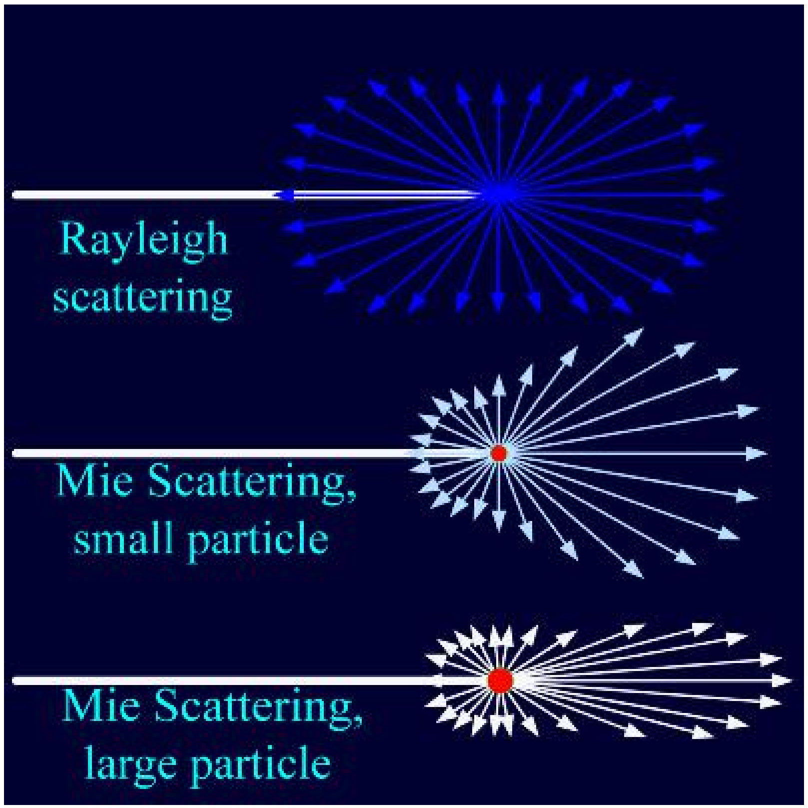
Passionate about Particulate
PCME

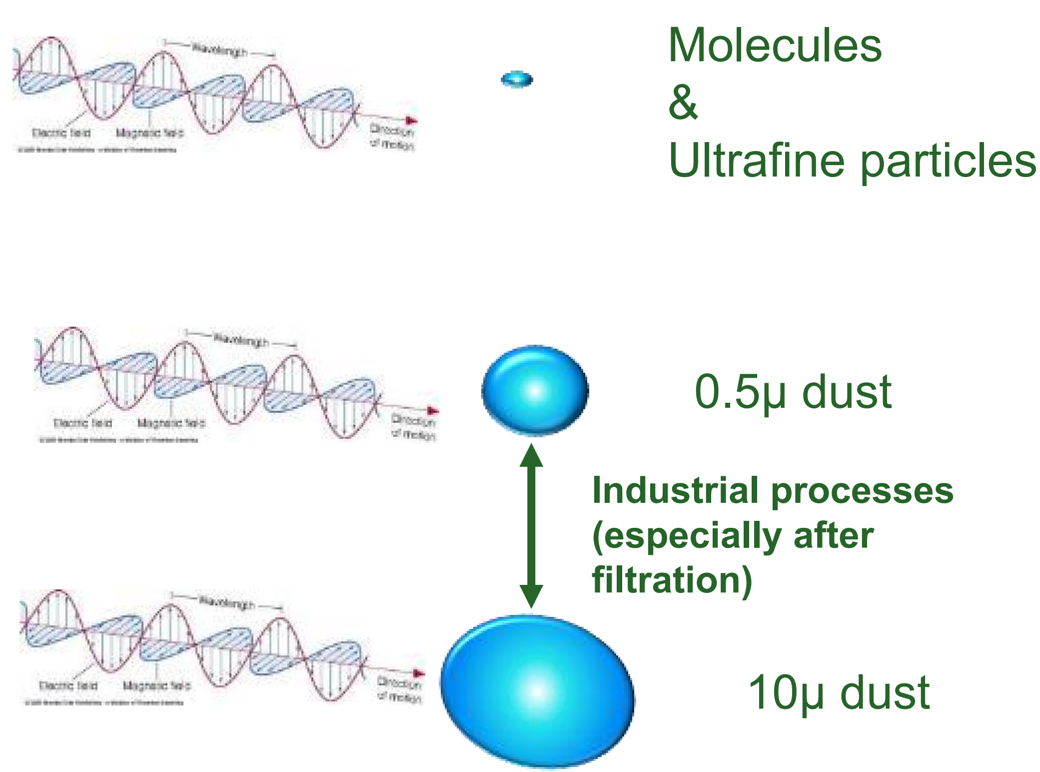
- ✓ Suitable measurement range of Particulate.
- ✓ What outputs you want from the system.
- ✓ Measurement Technology and Sampling-Continuous or Cyclic Sampling and Measurement.
- ✓ Process conditions such as Temperature, Velocity, Other Parameters present is the sampling stream corrosive due to Acid Gases such as HCL, HF etc.
- ✓ Sampling point- Diameter of stack, thickness of wall, length of flange, access and space on the platform to locate the monitor where you can get a representative sample. Please consider that Wet Stack Particulate Monitors are located at Platform level close coupled to the sampling point.
- ✓ Installation lifting and locating into position, size/footprint
- ✓ Utilities and Interface- Power, Air, Local interface (Display and Keyboard), Communication (RS232, RS485, Ethernet, 4/20mA, Relay outputs).
- ✓ Certification to comply with legislation and your permit? (European QAL1)
- ✓ Calibration Method 5 or 5B and Correlation Test Calculation (European EN14181 QAL2)
- ✓ Automatic built in Zero and Upscale checks to ensure stability and compliance to PS-11 (QAL3)
- ✓ Access and Serviceability
  - ✓ For initial set up and commissioning.
  - ✓ Please consider that extractive Particulate Monitoring systems will need to be cleaned from the sample probe or inlet through to the vent or return.
  - ✓ Serviced by site and external personnel.
  - ✓ Absolute Correlation Audit ACA for PS-11(EN14181 AST)
  - ✓ Diagnostic information.
- ✓ Budget initial purchase of equipment, installation and on going consumable parts.

Introducing Scatter technology for Dry and Humid applications

Passionate about Particulate PCME

Light is scattered in all directions





Largest scatter signal is from forward scatter so capable of measuring low concentrations typically found in many industrial applications!

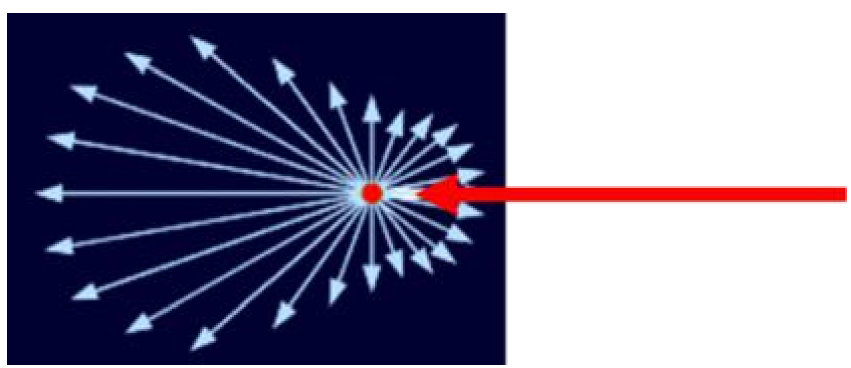
Introducing PCME QAL 181
ProScatter and Improved Forward Scatter Measurement Technique

Passionate about Particulate
PCME

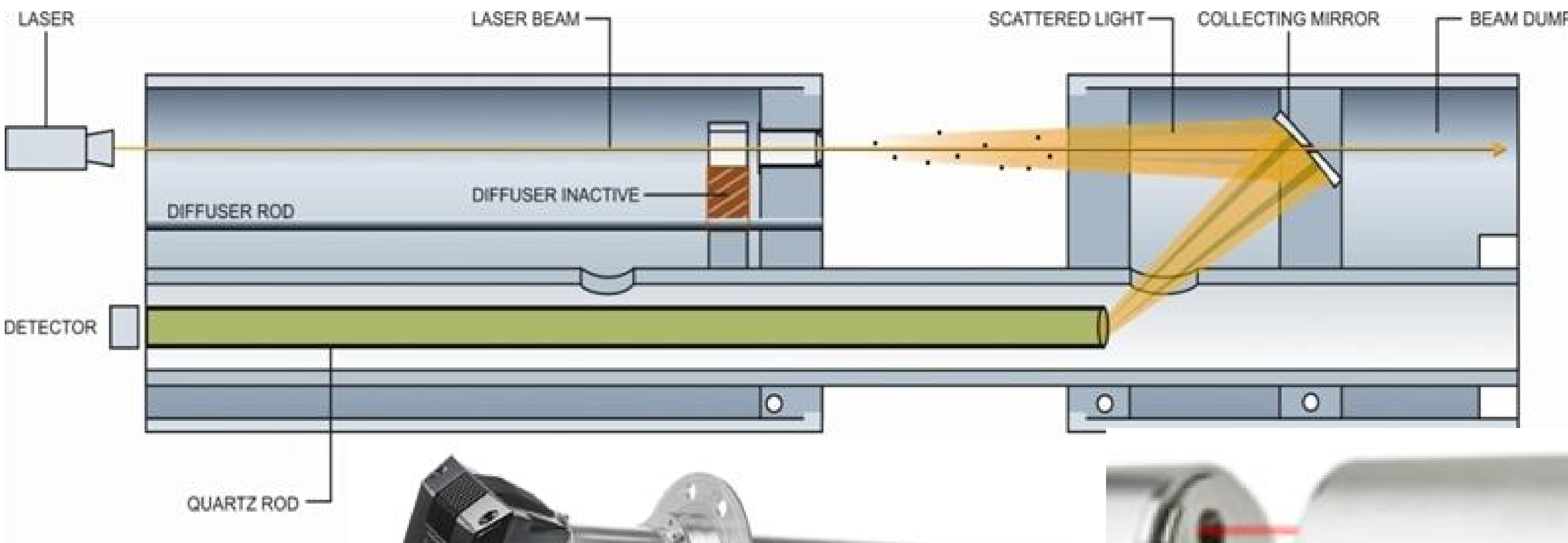
EN
15267-3
certified

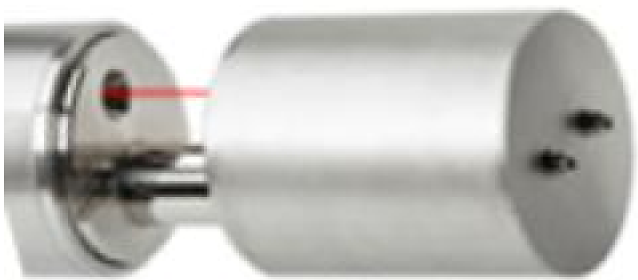
TÜVRheinland®
Precisely Right.

mCERTS
Sira MC090152/00



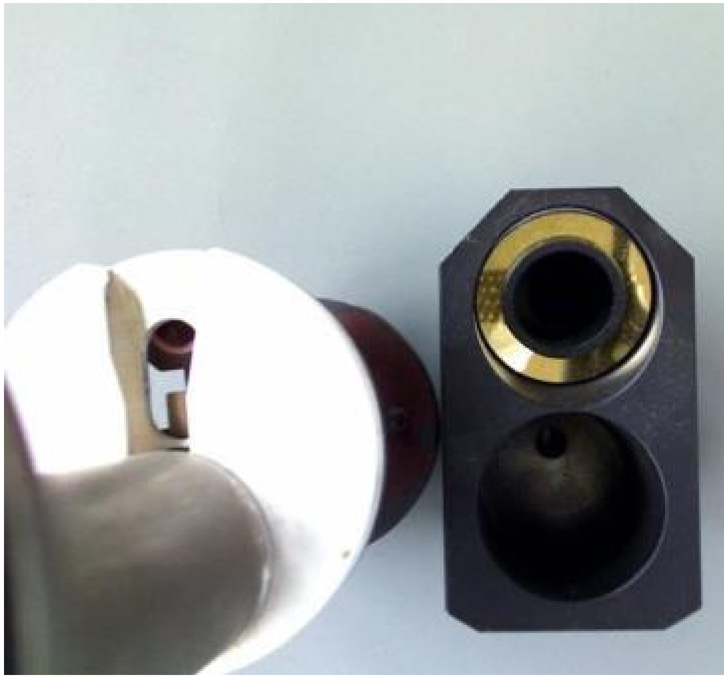
PCME QAL 181 ProScatter™
An Improved Forward Scatter Technology

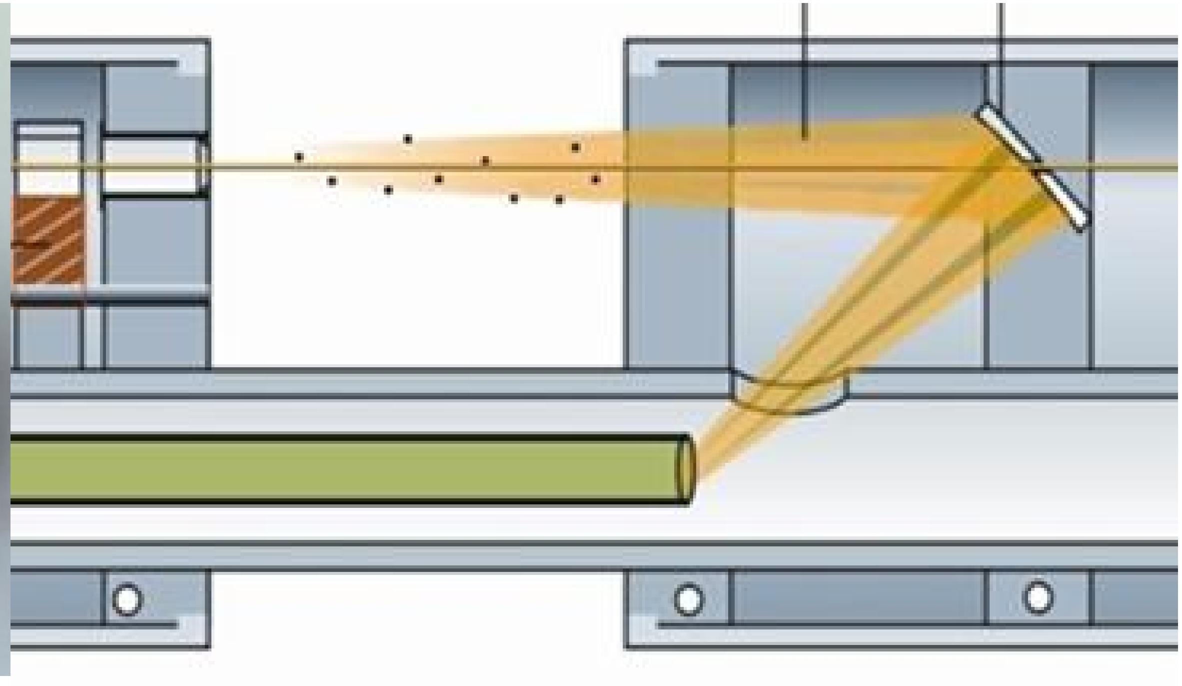




- ✓ PCME Stack181 The ideal solution for dry low dust applications.
- ✓ EN15267-3 QAL1 as defined by EN14181 with a certified range of 0-15mg/m³
- ✓ Automatic Zero and Span checks to satisfy QAL2.
- ✓ Manual Span and AST Linearity checks.
- ✓ Large dynamic measurement range and fast response.
- ✓ Easy to service on site.

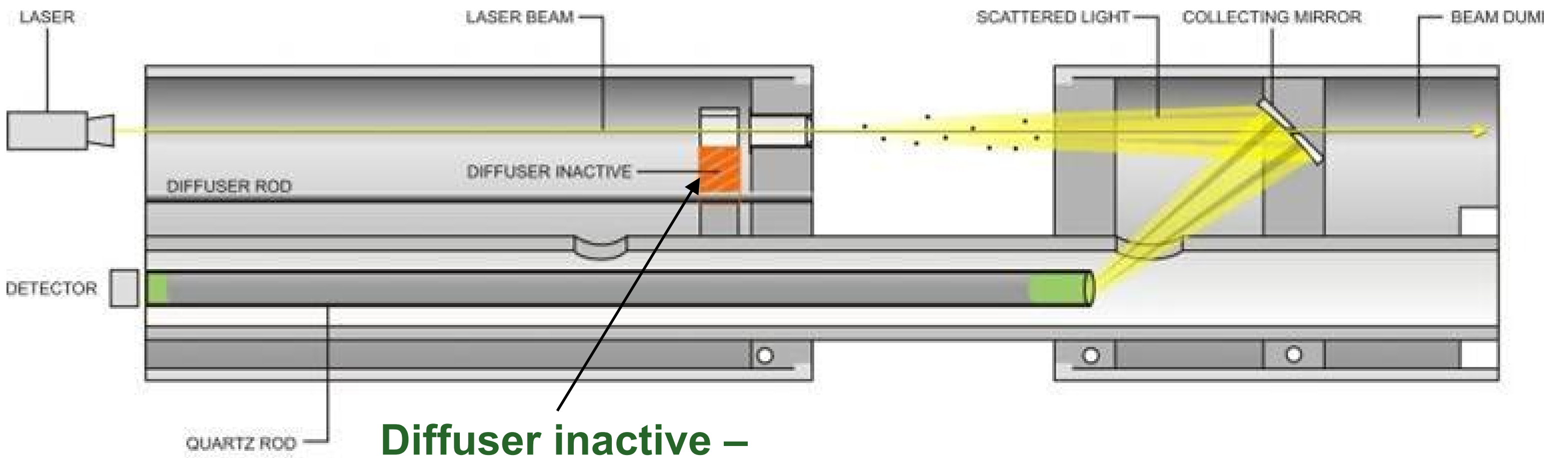
PCME QAL 181 ProScatter™
An Improved Forward Scatter Technology





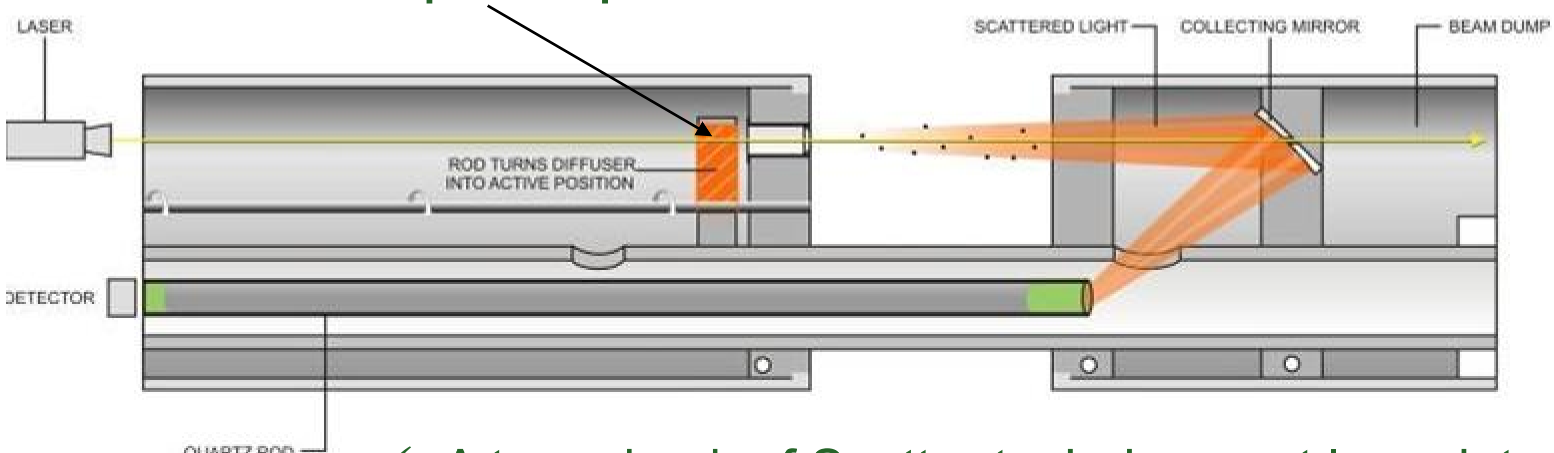
- ✓ Larger measurement volume
- ✓ Smaller angle of incidence meaning less impact due to particle size
- ✓ No moving parts in the measurement path.
- ✓ True check of forward scatter.
- ✓ Multi-lingual Text driven menu and display.
- ✓ Built in data loggers for added security of data.
- ✓ TCPIP Ethernet RS485 Modbus

PCME QAL 181- Automatic Zero & Span Checks to satisfy QAL3



Diffuser inactive – measurement position

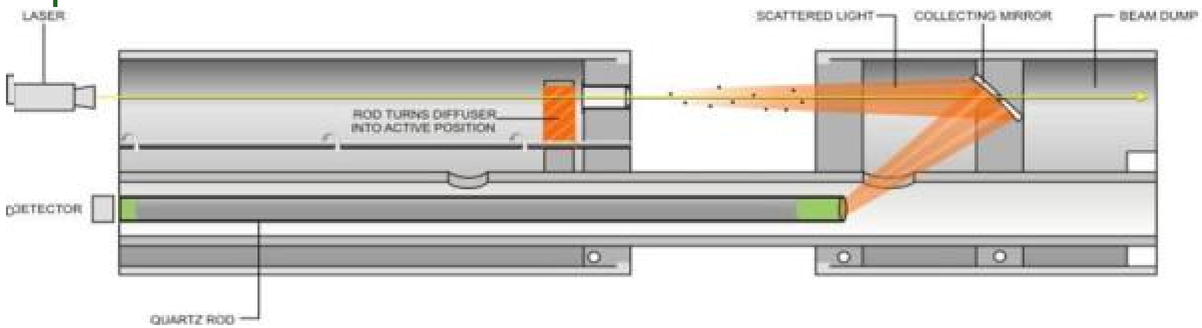
Diffuser active – Span test position



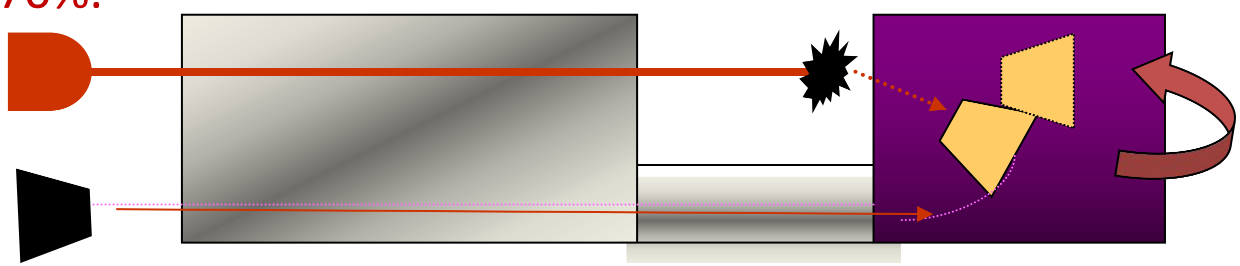
- ✓ A true check of Scatter technique not laser intensity!!

PCME advantages over other Forward Scatter based systems

✓ PCME perform a true Zero and Span check of the Forward Scatter measurement principle and the measurement probe without any moving parts in the measurement path.



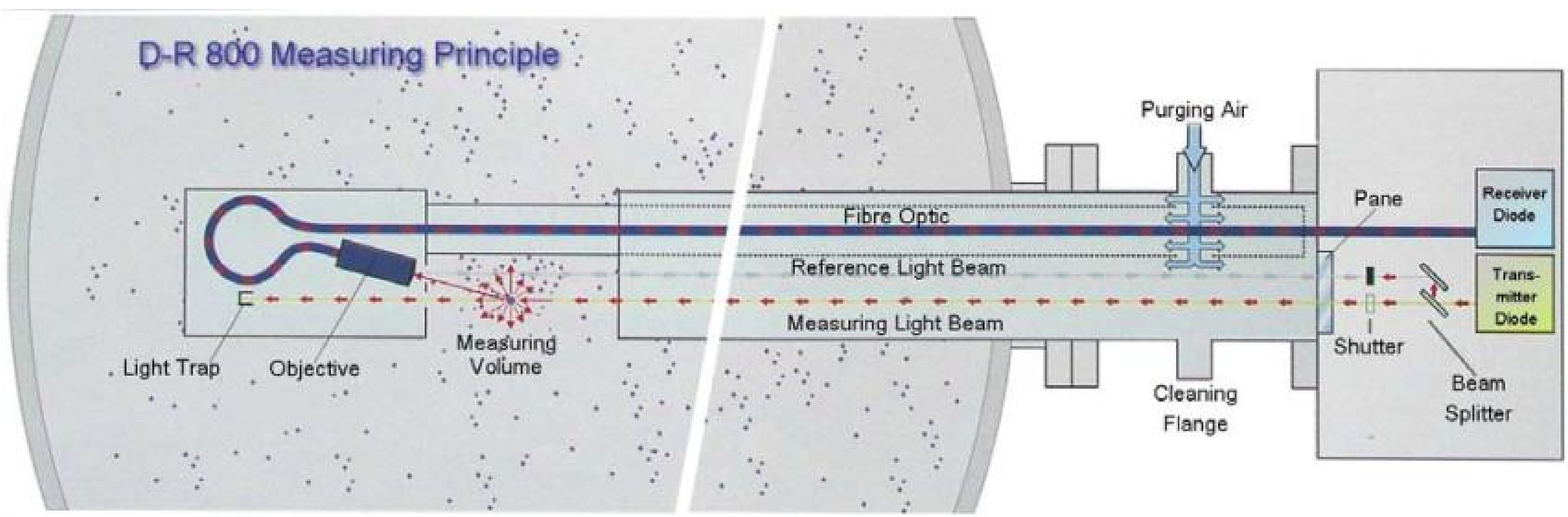
! Sick first move the detector to measure laser intensity then state in their manual that first the system performs a Contamination check at 100% laser intensity and performs a gain adjustment followed by a Span check where they reduce the laser intensity to 70%.



! Does this mean Sick make a gain adjustment compensating for any laser intensity drift? And does the Detector always return to the same position?

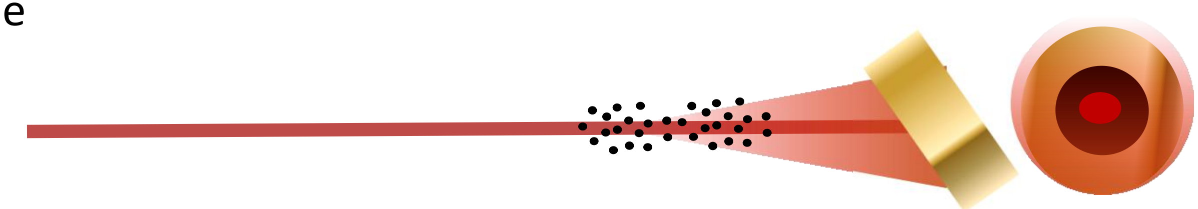
PCME advantages over other Forward Scatter based systems

- Durag/Dr Foedisch perform a Optical degradation/Pollution check every 5 minutes. **Why? And do they make some adjustment?**
- Durag/Dr Foedisch perform the span/laser intensity check by splitting the laser beam, this **is not** a forward scatter span check.

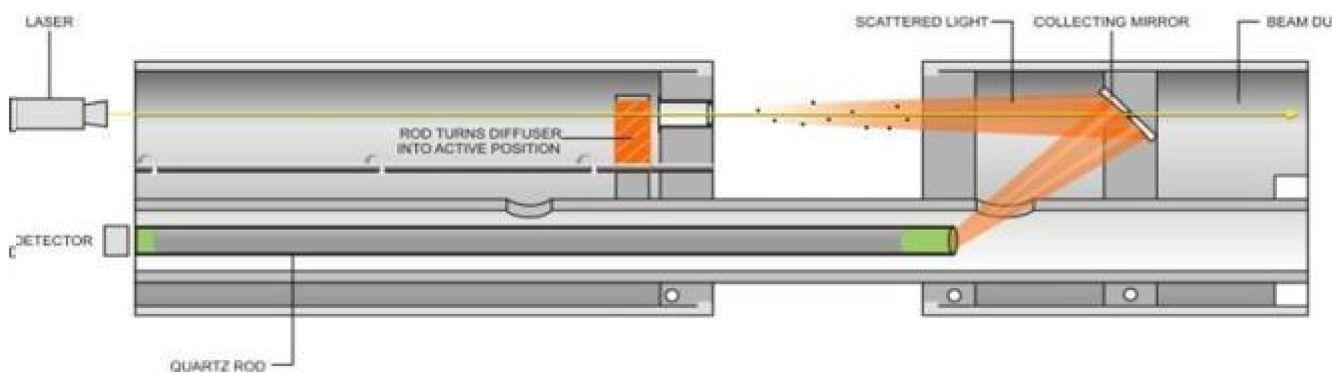


PCME QAL 181 ProScatter™
An Improved Forward Scatter Technology

- Conical mirror improves light collection by gathering full cone of scattered light 360° around the laser light source and is ~ 10x larger area of light collection.

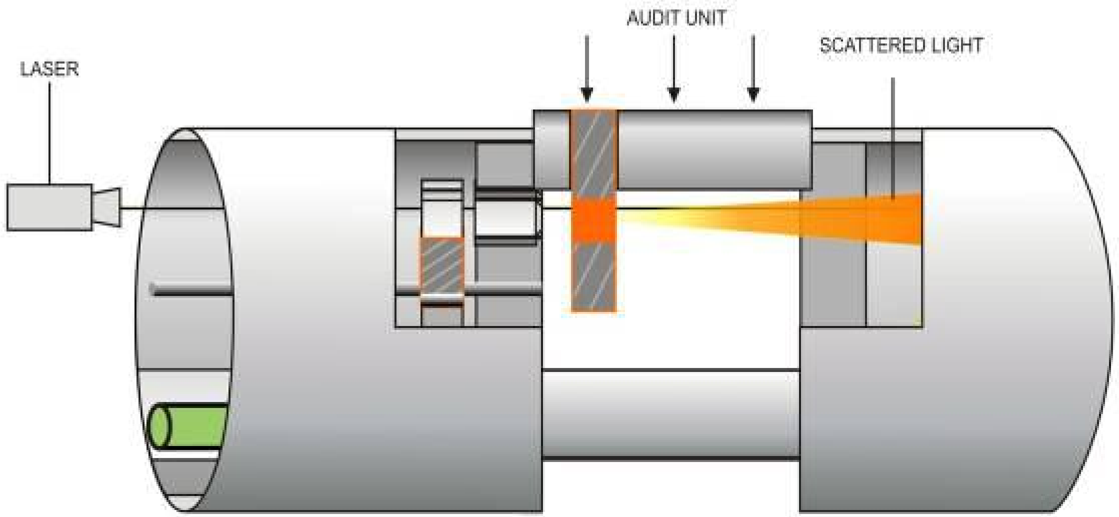


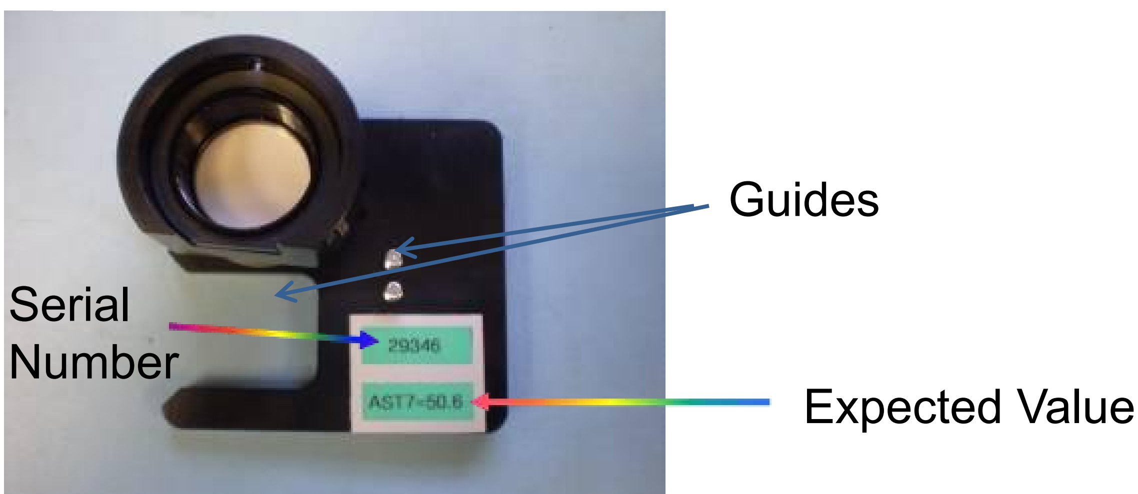
- Narrow forward scatter angle minimizes effect of changing particle size.
- While the Forward Scatter is still sensitive to changes in particle size, ProScatter has reduced sensitivity when compared to other Forward Scatter designs.



- Upscale check is provided by introducing a scatter body in light path to fully challenge the instruments ability to measure forward scattered light.
- Measurement optical path always remains in a fixed position for a reliable measurement.

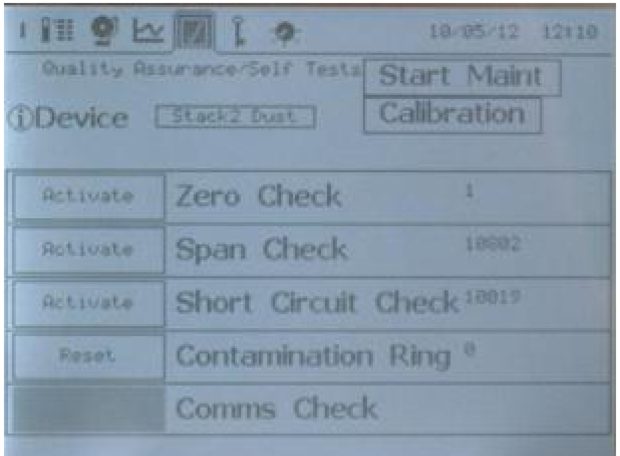
PCME QAL 181 External Audit Additional Span and AST Linearity





PCME QAL 181
Control, Display and Output

- ✓ Easy to use user interface
- ✓ Multi-lingual including Chinese
- ✓ Large graphics screen – see the process, QA, diagnostics
- ✓ Data logging
  - ✓ Long Term-15 minute average for CEM reporting.
  - ✓ Short Term – 1 minute average for Process monitori
  - ✓ Pulse Log – Instantaneous data for monitoring filter cleaning pulses.
  - ✓ Alarm log – Providing good diagnostic information.
- ✓ Set up, Display, Logging and Communication via outputs.



A single Particulate Monitor to provide a Stand alone and convenient measurement

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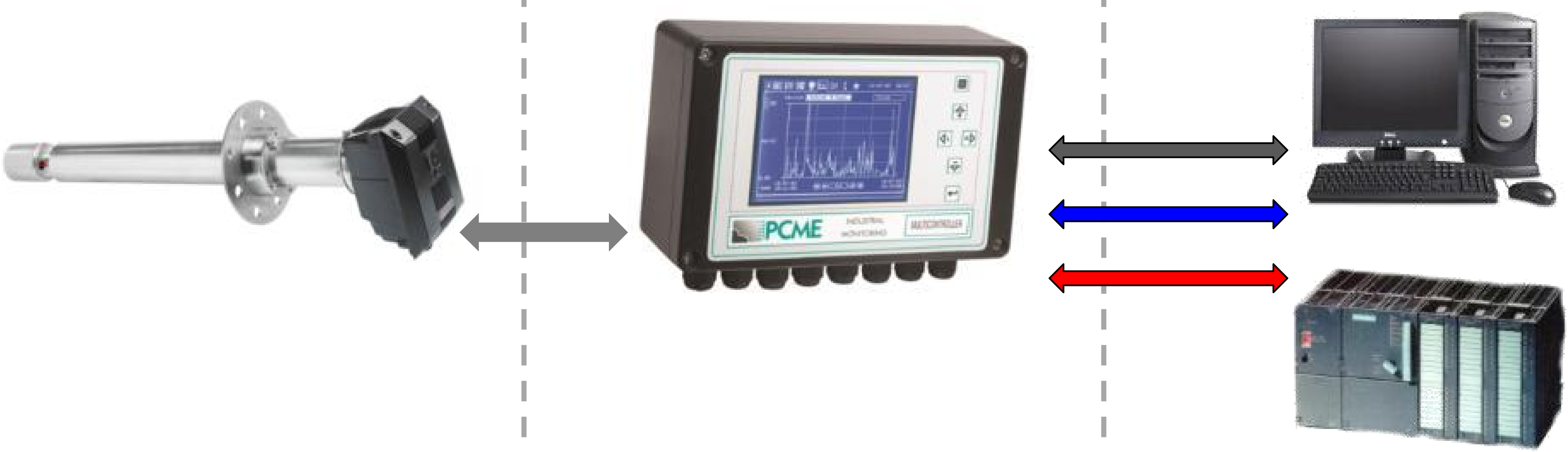
PCME

Stack

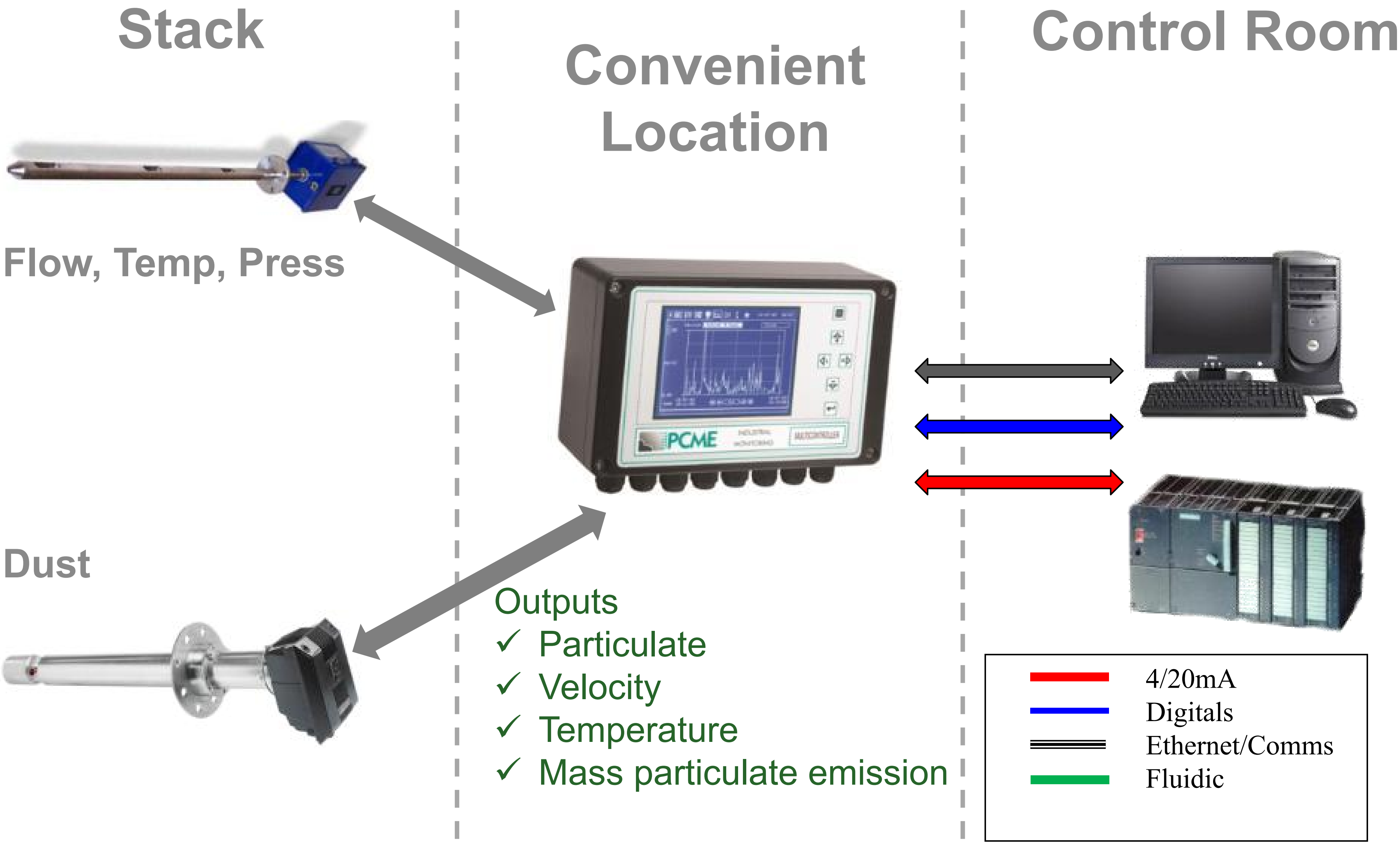
Convenient Location

Control Room



4/20mA
Digitals
Ethernet/Comms
Fluidic

Does your Particulate Monitor need to be providing a Mass measurement?



Wet Stack Particulate Monitoring

Previously there was no Extractive Particulate Monitoring System that could:

- Measure Particulate on a continuous basis!
- Or demonstrate it could meet the performance standards of EN14181 or PS-11 measuring low level Particulate concentrations typically found after modern abatement!!
- Or there was no complete system with the required type approval such as EN15267-3 with QAL1 as defined by EN14181 and with a suitably low certification range and dynamic measurement range!!!

- **Until now**
- **PCME are the first company to achieve EN15267-3 certification with QAL1 using the QAL181WS instrument.**

Introducing the PCME Wet Stack Particulate Monitoring System

Passionate about Particulate
PCME

- Continuous Direct Extractive Sampling.
- Continuous Measurement.
- Low Certification and dynamic measurement ranges.
- Two variants available:
    i. Complete Wet Stack System TUV Certified to EN15267-3 with QAL1 as defined by EN14181.
- Range of probe materials to suit process conditions.
- Easy to install and maintain on site.

Continuous & Direct Sampling System

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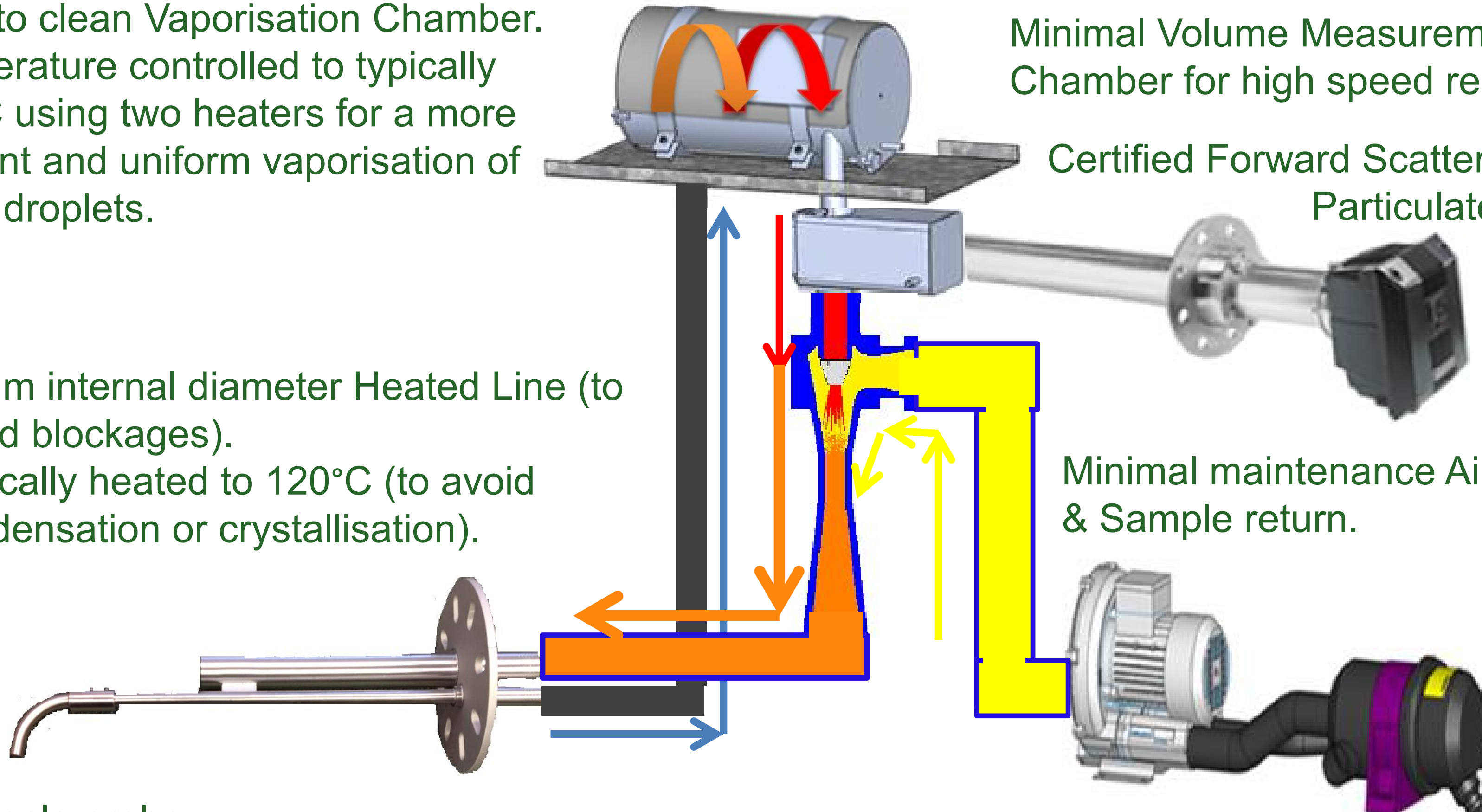
Easy to clean Vaporisation Chamber. Temperature controlled to typically 300°C using two heaters for a more efficient and uniform vaporisation of water droplets.

Minimal Volume Measurement Chamber for high speed response.

Certified Forward Scatter QAL181 Particulate Sensor.



30mm internal diameter Heated Line (to avoid blockages).
Typically heated to 120°C (to avoid condensation or crystallisation).

Minimal maintenance Air Ejector & Sample return.

Sample probe

Continuous sampling complete with sample return to stack to prevent contamination and safety issues.

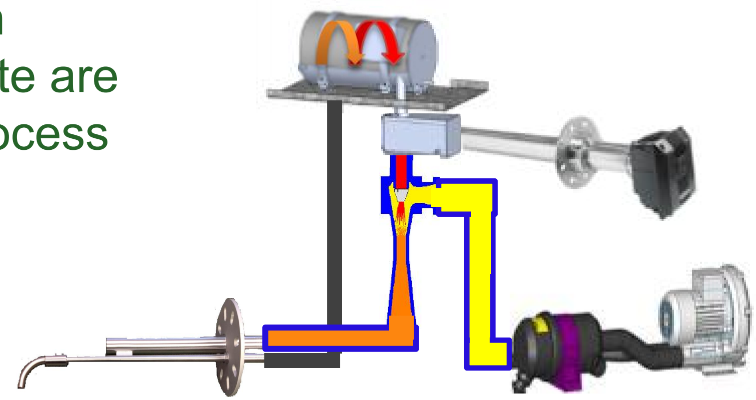
Variable speed Air Blower complete with Filter to obtain desired sampling velocity. No need for installation of expensive instrument air line or on going cost of providing large volume of instrument air.

Direct Sampling System versus Dilution

Benefits of Direct Sampling compared to Dilution

We know Emission Limit Values (ELV's) for Particulate are already quite low and depending on the industrial process they could be as low as 5mg/m3.

We also know that ELV's for Particulate are in many industries are being reduced.

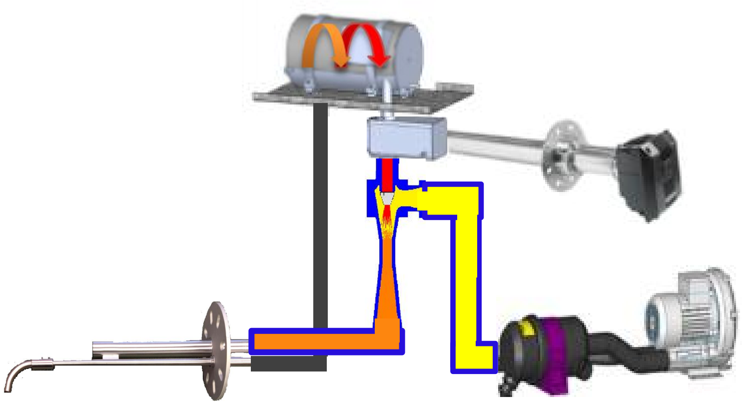


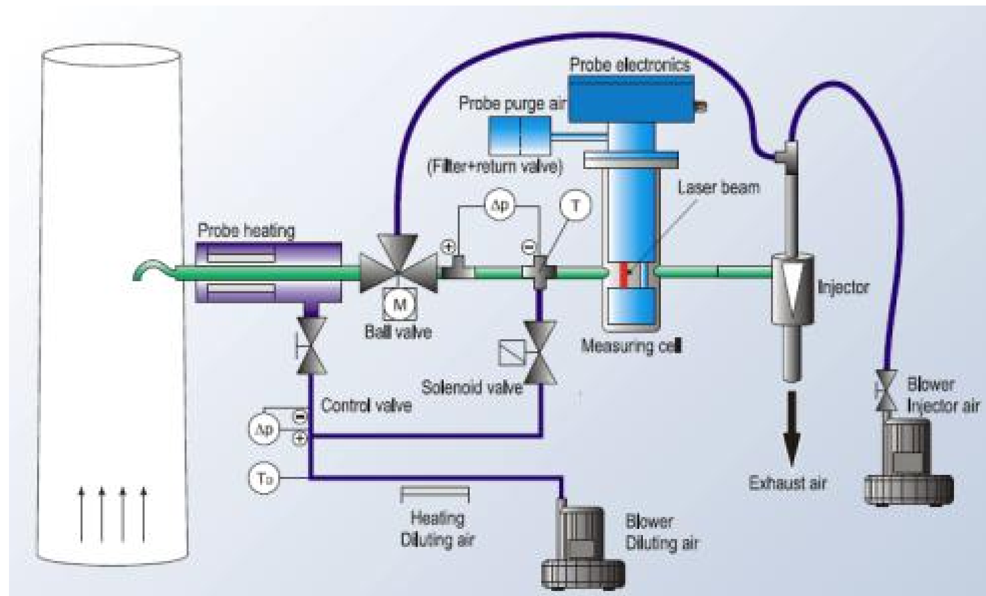
- ✓ Therefore for improved accuracy, resolution and effective calibration especially for low level Particulate measurement the whole sample and responding signal will need to be seen by the sensor to avoid significant measurement errors.
- x Using Dilution based sampling systems may work okay for gas measurement because you have end of probe filters to prevent particulate entering the sampling system, blocking critical orifice and altering sample flow rates which in turn changes the dilution ratio. Plus you can inject gas at the tip of the probe to determine and check dilution ratios which is used for calibration.
- x Unfortunately dilution ratios when sampling particulate cannot be determined by injection of dust at the probe. Dilution rations when sampling particulate will vary with varying size of particulate. Restrictions and blockages by particulate entering the sampling system will alter dilution ratios.
- x If we consider a level of particulate in the stack of between 2-5 mg/m3 with a 20:1 dilution ratio this will mean the dust monitor is trying to measure a sample of only **0.1 to 0.5 mg/m3!!!**
- x **Finally the air used for dilution needs to be dust free using filters does not guarantee dust free air only the maximum particulate size, this adds to the measured errors!!!!!!!!!!**

PCME advantages over other Forward Scatter based systems

Passionate about Particulate PCME

- ✓ Direct extractive system avoids all the errors associated with Dilution based systems such as Durag/Dr Foedisch.



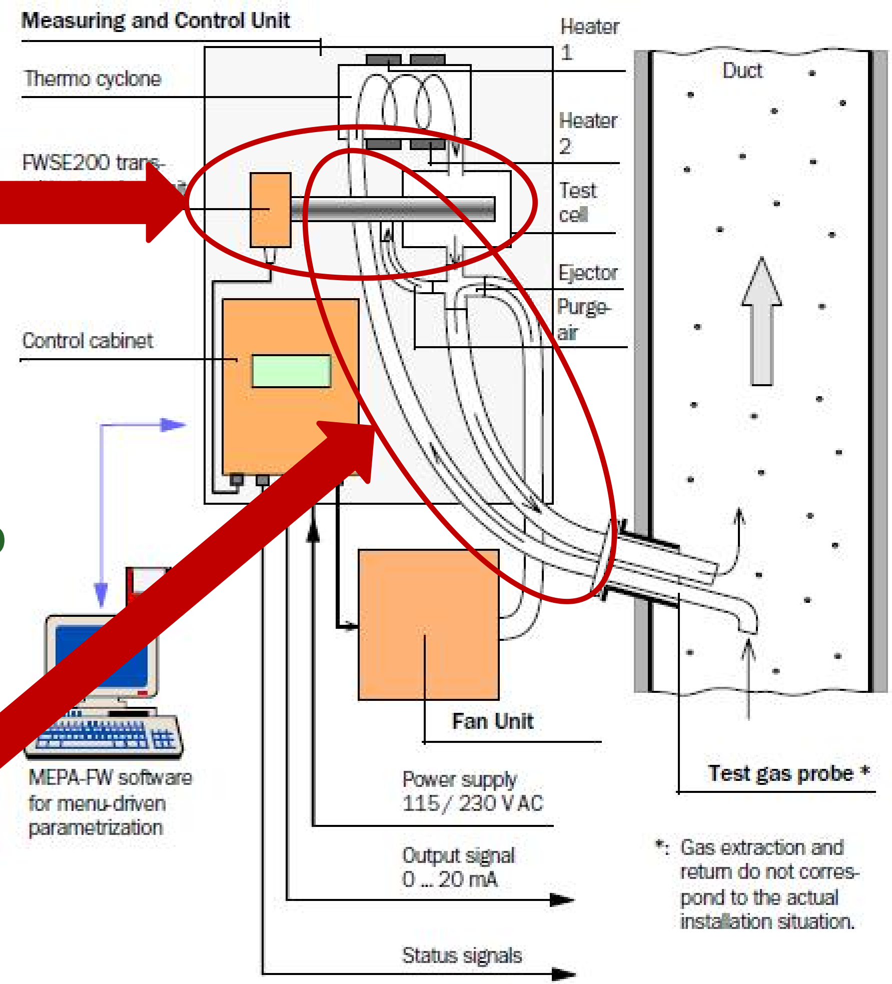


! Durag/Dr Foedisch performs an optical pollution check every 5 minutes. **Why?**

! Durag/Dr Foedisch is the same system it is a collaboration between two companies one doing the sampling the other doing the measurement!

PCME advantages over other Forward Scatter based systems

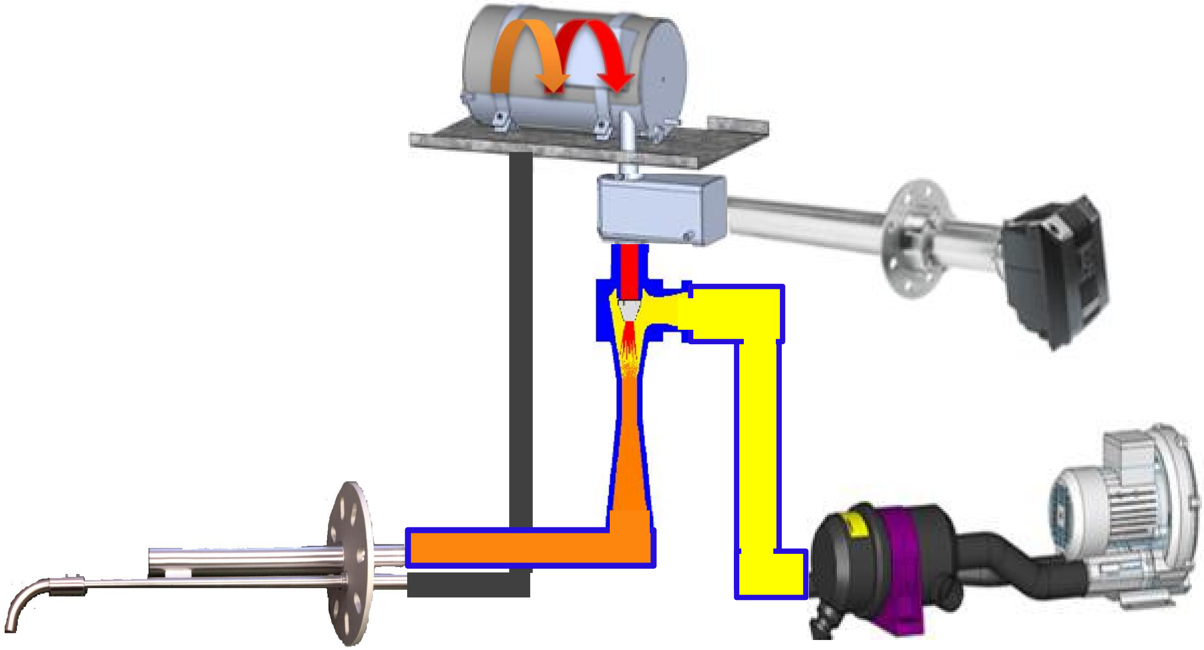
- ✓ In Europe PCME have the complete system certified to EN15267-3.
- ! Please note that only the Sensor is certified within the Sick System.
- ! The complete system needs to be certified to EN15267-3 to fully satisfy EN14181.



- ✓ PCME use Heated Sample inlet tube to avoid condensation and cold spots which can also create a build up and sample restrictions
- ! Sick sample inlet is not heated

PCME advantages over other Forward Scatter based systems

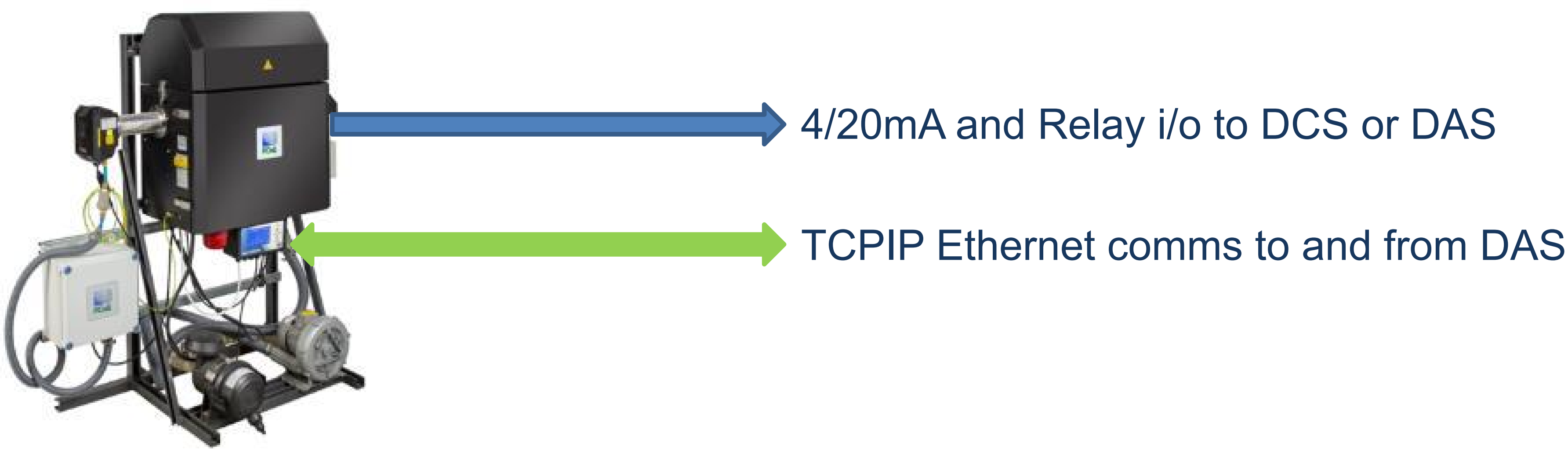
- ✓ The PCME QAL 181-WS (European variant) or PCME Stack 181-WS US EPA Variant for PS-11) can sample at a fixed velocity or with the aid of an external flow meter supplying a 4/20mA input signal, PCME simply increase the speed of the blower until the measured sample velocity matches the process velocity to achieve Isokinetic sampling.



! Durag/Dr Foedisch or the Sick systems use fixed sampling velocity.

PCME advantages over other Forward Scatter based systems

- ✓ PCME provide multiple 4/20mA signals, for example one 4/20mA signal can represent the particulate concentration and another 4/20mA the actual sample velocity which can be used as an indicator of sampling blockages!
- ✓ In addition PCME also provide RS232, RS485 Modbus and Ethernet communication-Ideal for carrying measurement and diagnostic information to Data Acquisition Systems



! Durag/Dr Foedisch supply 4/20mA and relay I/O

! Sick supply both RS232 plus 4/20mA and Relay I/O

More PCME advantages over other systems

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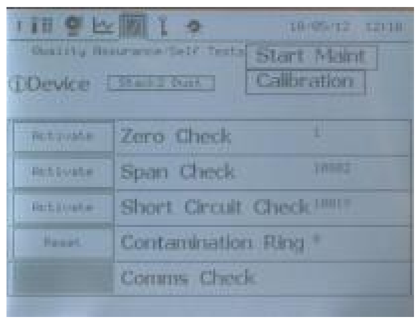
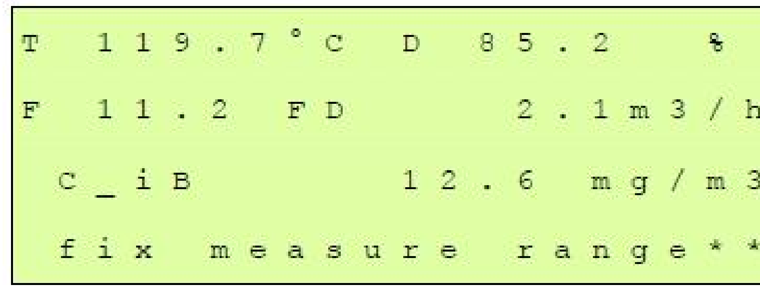
✓ The PCME system uses a controller complete with data logging with the menu driven display based on easy to follow multilingual text.

! Durag/Dr Foedisch and Sick systems use difficult to understand English or German code/text (not easy to follow text characters).





✓ Finally please remember two last points:

1. Beta Gauge is not a continuous measurement it is a cyclic measurement with typically 1 measurement every 20-30 minutes so it cannot form valid half hourly measurements.

By using some of these sales arguments you have many seeds of doubt that you can drop into conversation.

Wet Stack example.

EPRI Lead Trial Demonstrating PS11 compliance

- Industry: Detroit Thermoelectric- Monroe
- Contractor, EPRI
- Application: Coal fired Power plant
- Filtration System: FGD with SCR
- Temperature: 54^{o}C
- Stack diameter: 8.7m
- Dust level: 2 – 4 mg/m^3
- Moisture content: 15% (saturated) + 12%
- Past Equipment used: none
- Length of Trial: 6 months

✓ Low concentration calibration R2 correlation with Isokinetic sampling

✓ 4 week routine maintenance required

- Probe cleaning (dust)
- Heater cleaning (crystal deposits)

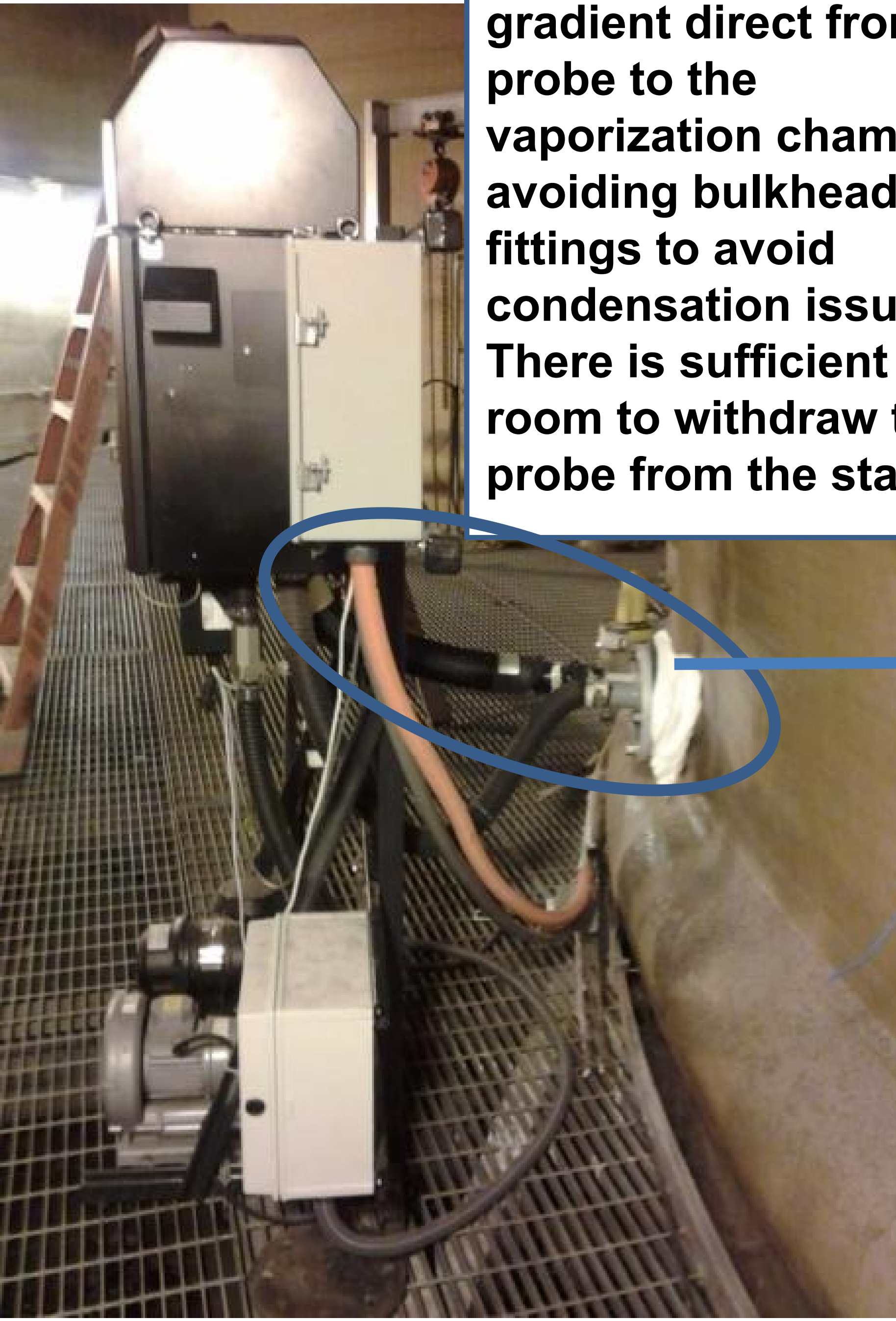
✓ **Successful PS-11 trial unlike other suppliers!!!**

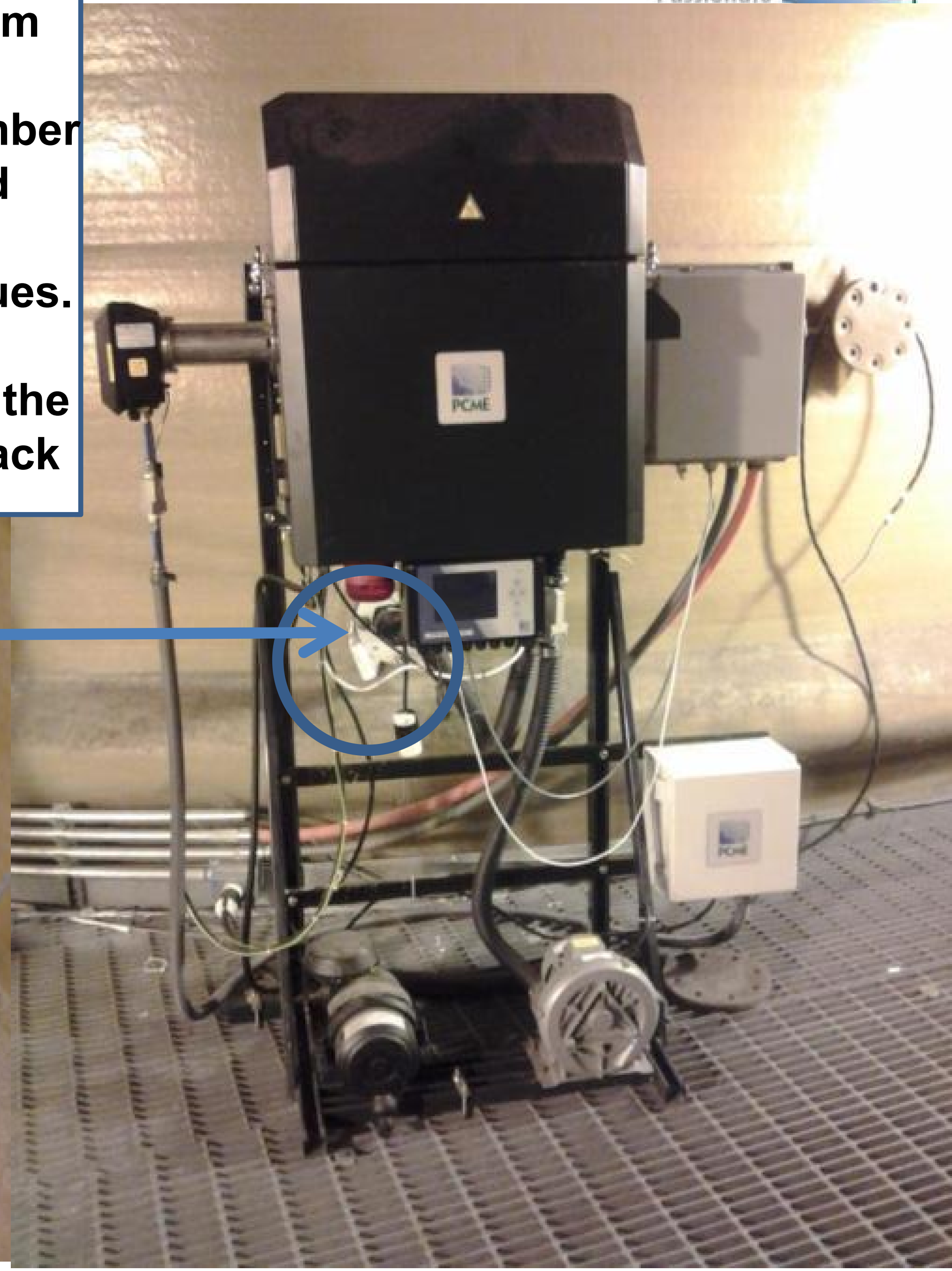
Lock out Specification

- ✓ Must satisfy US EPA PS-11.
- ✓ Continuous and Direct Heated Sampling of particulate for continuous measurement for the formation of valid 1 minute, 15 Minute, Half hourly and hourly averages within the chosen DAS. (Heated sampling to avoid further condensation causing blockages in colder weather).
- ✓ No moving parts in measurement path. (for Reliability and reducing failures).
- ✓ Variable Isokinetic Sampling capability, plus fixed automatic sampling velocity capability.
- ✓ Forward Scatter measurement technique including zero and span check that fully challenges measurement technique and complete measurement path.
- ✓ No compensations or adjustment of gains during zero and span drift checks are permitted. (The avoid step changes in measured values).
- ✓ TCPIP Ethernet or RS485 communication to include measurement and diagnostic information. (for longer communication runs)
- ✓ Minimum of 4x 4/20ma, 4 x Relay o/p, 1 x Relay i/p.
- ✓ Configuration, Display of Measurement and Diagnostics information via multi-lingual text driven display. (No external proprietary manufacturers software required with all settings saved in case of power failure or disconnection should power up and run without operator intervention)

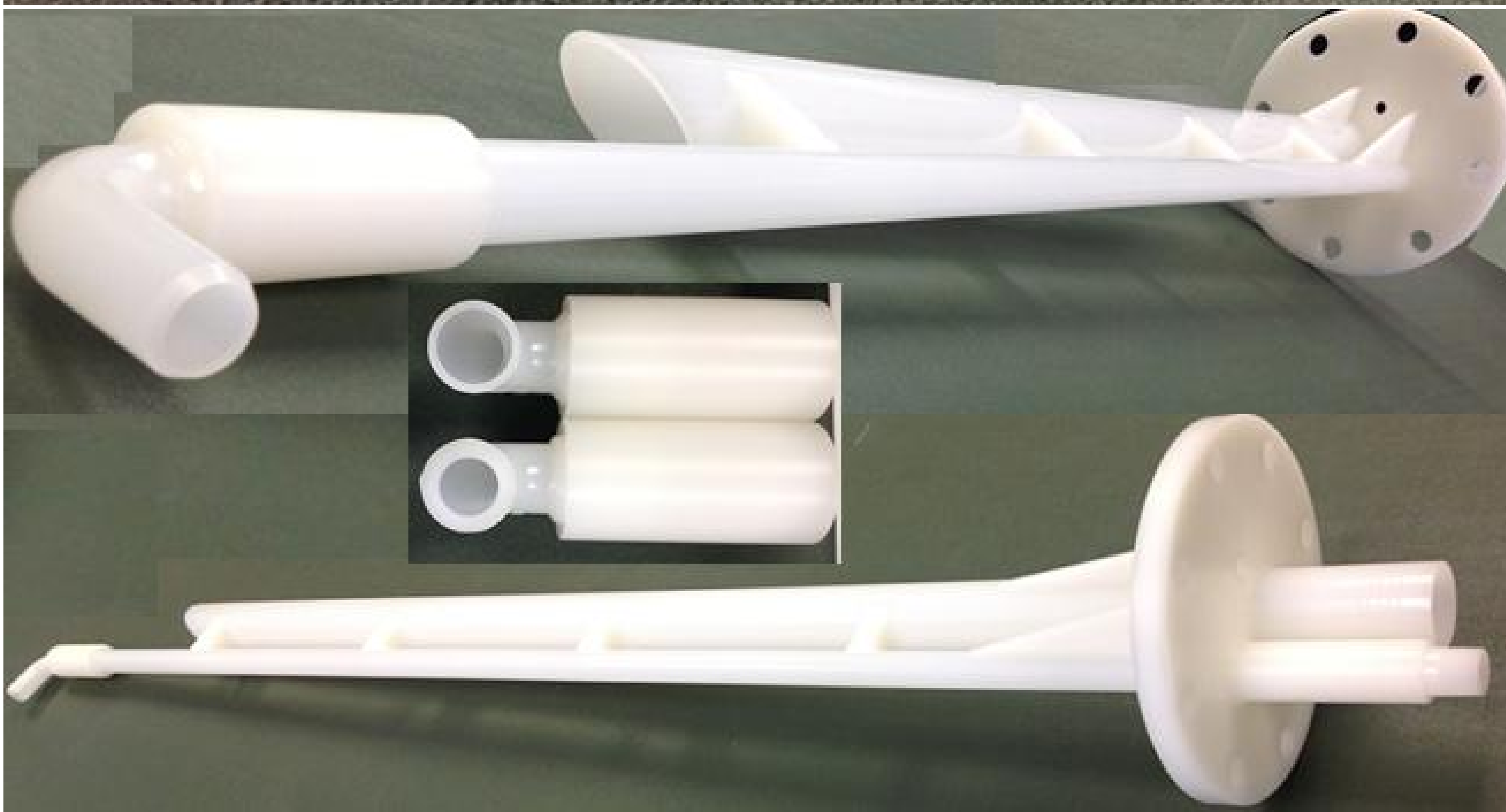
Passionate

Heated line at a slight gradient direct from probe to the vaporization chamber avoiding bulkhead fittings to avoid condensation issues. There is sufficient room to withdraw the probe from the stack





CME



Why PCME as your Partner?

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- ✓ Particulate monitoring is our core business.
- ✓ We have a choice from a range of measurement technologies to best suit the application.
- ✓ Our product range offer a range of value added features and options.
- ✓ Measurement, data logging, data acquisition and report software from a single company. One point of responsibility.
- ✓ Quality product and services from an ISO9001 and ISO14001 company.
- ✓ Continued investment in Research and Development to provide the best technologies to you. Including various patents.
- ✓ Proven track record with many successful independent industry lead trials.
- ✓ We have various type approvals and certification to suit the application.
- ✓ installation base of 30,000+ systems and 23 years industry experience.
- ✓ Service and Support – Providing long term support to ensure longevity of your investment and a long term relationship.

Thank You!

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THE QUEEN'S AWARDS FOR ENTERPRISE INTERNATIONAL TRADE 2013

Any Questions?